



$$\bar{C}^{[0]} \hat{U}^{[1]} = |1\rangle\langle 1| \otimes \hat{I} + |0\rangle\langle 0| \otimes \hat{U}$$

$$= \hat{I} \otimes \hat{I} + |0\rangle\langle 0| \otimes (\hat{U} - \hat{I})$$